Nonreduced orthogonal nine-frames in dimension eight

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Abstract

An open-ended autonomous run of Recursive Reasoning Exploration (RRE) generated a counterexample to the conjecture that the ideal of pairwise orthogonality relations is radical for every number of vectors and every ambient dimension. For nine vectors in dimension eight, this ideal is nonradical over every field of characteristic different from two, for every nondegenerate symmetric bilinear form. An explicit open chart reduces the claim to ten cubic equations in twenty-four variables. A 36-term polynomial represents a nonzero square-zero class. An integral ideal-membership identity and an integral linear functional certify the two assertions, so the arithmetic verification works over every field, including characteristic two, without a Gröbner-basis computation. This all-characteristic conclusion concerns the auxiliary diagonal-parameter scheme; its identification with an open frame chart requires characteristic different from two. We provide the certificate and a short checker using only Python's standard library. Stabilization gives nonreducedness whenever $d \geq 8$ and $n \geq d+1$. An accompanying three-view visualization guides the reader through the discovery route, proof dependencies, and provenance of the mathematical inputs.

1 The counterexample

For the frame statements, let K be a field of characteristic different from two, let (E, b) be a nondegenerate symmetric bilinear space of dimension d, and write $\mathbf{v}_1, \ldots, \mathbf{v}_n$ for generic vectors in E. Set

$$I_b(d,n) = (b(\mathbf{v}_i, \mathbf{v}_j) : 1 \leq i < j \leq n), \qquad X_b(d,n) = \operatorname{Spec} K[E^{\oplus n}]/I_b(d,n).$$

For the standard dot product we omit the subscript b. Conca and Welker conjectured that the off-diagonal Gram ideal is radical for all sizes, at least in characteristic zero [1, Conjecture 9.6]. Casabella and Sammartano restate the conjecture for orthogonal frames [2, Conjecture 6]. We prove the following negative answer.

Theorem 1. *For every field K of characteristic different from two and every nondegenerate symmetric bilinear form b on K^8, the ideal $I_b(8,9)$ is not radical.*

In particular, the theorem applies to the usual sum-of-products equations over $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. The statement concerns radicality as an ideal, independently of the set of real points. We do not assert that these parameters are minimal; Section 4.1 records the known and unresolved parameter ranges.

The proof consists of a small integral certificate and an explicit scheme isomorphism. The certificate establishes the crucial ideal membership and nonmembership by integer arithmetic. The chart argument transfers the result to the original quadratic ideal. No small-dimensional radicality result or unproved induction hypothesis enters the proof.

2 An integral square-zero certificate

In this section the ground field may have any characteristic, including characteristic two. Consider

$$R_{\mathbb{Z}} = \mathbb{Z}[q_1, q_2, q_3, q_4, c_{ai} : 1 \le a \le 4,\ 1 \le i \le 5], \quad F_{ij} = \sum_{a=1}^{4} q_a c_{ai} c_{aj}, \quad J = (F_{ij} : i < j).$$

We specify the witness first. Put

$$a_i = c_{1i}, \quad b_i = c_{2i}, \quad d_i = c_{3i}, \quad x_i = c_{4i}, \qquad [ij] = b_i d_j - b_j d_i.$$

Every cyclic sum in the following formula runs over the three triples $(1,2,3)$, $(2,3,1)$ and $(3,1,2)$ for (i,j,k). Define

$$\begin{aligned} h = x_5\Big\{ & b_4 \sum_{\mathrm{cyc}(i,j,k)} a_j a_k x_i^2 \big(x_k[j5] - x_j[k5] - x_5[jk]\big) \\ & + x_4 x_5 \sum_{\mathrm{cyc}(i,j,k)} a_j a_k b_i x_i [jk] \\ & - x_5 \sum_{\mathrm{cyc}(i,j,k)} a_k (a_i b_j - a_j b_i)[k4] x_i x_j \Big\}, \qquad g = q_4 h. \end{aligned} \tag{1}$$

The expanded h has 36 terms and total degree nine in $C = (c_{ai})$. Its column degrees are $(2,2,2,1,2)$.

Lemma 2 (Integral certificate)**.** *There are polynomials $P_{ij} \in R_{\mathbb{Z}}$ and a $\mathbb{Z}$-linear functional $\lambda : R_{\mathbb{Z}} \to \mathbb{Z}$ of finite monomial support such that*

$$q_4 g = \sum_{i<j} P_{ij} F_{ij}, \qquad \lambda(J) = 0, \qquad \lambda(g) = 1. \tag{2}$$

Consequently, for every field K, the image of g in $R_{\mathbb{Z}} \otimes K / J_K$ is nonzero and has square zero.

Computer-assisted verification. The accompanying file

`diagonal_integral_certificate.json`

lists all coefficients and exponent vectors of g, all monomial terms of the P_{ij}, and all nonzero values of λ on monomials. The variable order is column first:

$$c_{11}, c_{21}, c_{31}, c_{41}, c_{12}, \ldots, c_{45}, q_1, q_2, q_3, q_4.$$

There are 552 terms across the ten P_{ij} and 124 monomials in the support of λ. The integer coefficients in both lists have absolute value at most three. The supplied checker first expands (1) and verifies that it equals the listed g. It then expands the right-hand side of (2) and verifies exact equality with $q_4 g$ over $\mathbb{Z}$.

For completeness, the nonmembership verification is exhaustive without any degree bound. For each supported monomial M of λ and each term $T = q_a c_{ai} c_{aj}$ of F_{ij} dividing M, add $\lambda(M)$ to the entry indexed by $(i, j, M/T)$. For an arbitrary monomial U, the resulting entry is precisely $\lambda(UF_{ij})$. Exactly 144 entries occur, and all are zero. Every other UF_{ij} pairs trivially with λ, since none of its terms belongs to the support. Thus λ annihilates every monomial multiple of every generator, and hence the entire ideal J. Direct pairing gives $\lambda(g) = 1$.

These are integer identities. After specialization to any field, the functional still annihilates the ideal and takes value one on g, so $g \notin J_K$. Finally,

$$g^2 = h(q_4 g) = h \sum_{i<j} P_{ij} F_{ij} \in J.$$

The checker also verifies this polynomial identity directly. The verification uses no output from the discovery engine as an assumed generator or an assumed normal form. □

The finite certificate is part of the proof. Section 5 describes its format and reproduction. In particular, the proof does not infer a characteristic-zero statement from a modular experiment.

Characteristic scope. Lemma 2 includes characteristic two. Its nonreducedness conclusion concerns the auxiliary diagonal-parameter scheme $\operatorname{Spec} K[q, C]/J(4, 5)$ over every field. This does not assert a characteristic-two frame theorem: the chart identification in Lemma 3 uses $H = A + \operatorname{diag}(q)/2$ and requires 2 to be invertible. Accordingly, Theorem 1 and Corollary 4 apply only in characteristic different from two.

3 The open chart and the frame theorem

For the remainder of the paper, assume $\operatorname{char} K \neq 2$. We give the chart in arbitrary even dimension. Equip $K^r \oplus K^r$ with the split form

$$b((x, y), (x', y')) = x^{\mathsf{T}} y' + y^{\mathsf{T}} x'.$$

For an $r \times m$ matrix C, set

$$J(r, m) = \left(\sum_{a=1}^{r} q_a c_{ai} c_{aj} : 1 \leq i < j \leq m \right).$$

Lemma 3. *On the open subset of $X_b(2r, r+m)$ where the upper $r \times r$ block of the first r columns is invertible, there is an isomorphism*

$$X_b(2r, r+m)\big|_{\det L \neq 0} \simeq \mathrm{GL}_r \times \mathbb{A}^{r(r-1)/2} \times \operatorname{Spec} K[q_a, c_{ai}]/J(r, m). \tag{3}$$

Proof. We give both maps on points over an arbitrary commutative K-algebra Λ, including a nonreduced one. Write a frame in block form as

$$M = \begin{bmatrix} L & P \\ B & Z \end{bmatrix}, \qquad L, B \in \mathrm{Mat}_{r\times r}(\Lambda), \quad P, Z \in \mathrm{Mat}_{r\times m}(\Lambda),$$

and suppose $\Delta = \det L$ is a unit. Its equations are

$$\begin{aligned} &\operatorname{offdiag}(L^{\mathsf{T}} B + B^{\mathsf{T}} L) = 0, \\ &L^{\mathsf{T}} Z + B^{\mathsf{T}} P = 0, \\ &\operatorname{offdiag}(P^{\mathsf{T}} Z + Z^{\mathsf{T}} P) = 0. \end{aligned} \tag{4}$$

Here offdiag denotes the entries indexed by $i < j$; there are no equations on the diagonal pairings.

Forward map. Set

$$H = L^{\mathsf{T}} B, \qquad C = L^{-1} P, \qquad D = L^{\mathsf{T}} Z.$$

The first block of equations gives $H + H^{\mathsf{T}} = Q$, where $Q = \mathrm{diag}(q_1, \dots, q_r)$ and $q_i = 2H_{ii}$. Thus $A = H - Q/2$ satisfies $A^{\mathsf{T}} = -A$ and $A_{ii} = 0$. The mixed equations give

$$D + H^{\mathsf{T}}C = L^{\mathsf{T}}Z + B^{\mathsf{T}}LL^{-1}P = 0.$$

Consequently $D = -H^{\mathsf{T}}C$. The last block of equations becomes

$$\mathrm{offdiag}(C^{\mathsf{T}}D + D^{\mathsf{T}}C) = -\,\mathrm{offdiag}(C^{\mathsf{T}}QC) = 0.$$

The forward map is therefore $M \mapsto (L, A, q, C)$.

Inverse map. Given (L, A, q, C) with L invertible, $A^{\mathsf{T}} = -A$, $A_{ii} = 0$, and $J(r, m) = 0$, put $H = A + Q/2$ and set

$$\Psi(L, A, q, C) = \begin{bmatrix} L & LC \\ L^{-\mathsf{T}}H & -L^{-\mathsf{T}}H^{\mathsf{T}}C \end{bmatrix}. \tag{5}$$

This matrix satisfies all original equations. Its first-block pairing matrix is $H + H^{\mathsf{T}} = Q$, hence diagonal; its mixed pairing matrix is $-H^{\mathsf{T}}C + H^{\mathsf{T}}C = 0$; its last-block pairing matrix is $-C^{\mathsf{T}}(H + H^{\mathsf{T}})C = -C^{\mathsf{T}}QC$, whose off-diagonal entries vanish by $J(r, m)$.

The two compositions. Starting with a frame M, the reconstructed blocks satisfy

$$B' = L^{-\mathsf{T}}L^{\mathsf{T}}B = B, \qquad P' = LL^{-1}P = P.$$

For the lower-right block the mixed equations are essential:

$$Z' = -L^{-\mathsf{T}}H^{\mathsf{T}}C = L^{-\mathsf{T}}D = L^{-\mathsf{T}}L^{\mathsf{T}}Z = Z.$$

The upper-left block remains L. Conversely, starting with parameters and applying the forward map to (5) gives

$$H' = L^{\mathsf{T}}L^{-\mathsf{T}}H = H, \qquad C' = L^{-1}LC = C, \qquad q_i' = 2H_{ii} = q_i, \qquad A' = H - Q/2 = A.$$

Both compositions are identities. Entries of L^{-1} are cofactors divided by Δ, so all formulas are regular on the indicated open subscheme. They commute with arbitrary changes of Λ and give inverse scheme morphisms, preserving the full nilpotent structure. □

Proof of Theorem 1. First suppose b is split, and apply Lemma 3 with $r = 4, m = 5$. To make nonvanishing precise, write

$$R_0 = K[q_a, c_{ai}]/J(4, 5), \qquad R_{\mathrm{chart}} = R_0[\ell_{ij}, a_{ij}\ (i < j), u]/(u\det(\ell) - 1),$$

where the ℓ_{ij} range over all sixteen matrix entries and the six a_{ij} are the independent entries of the alternating matrix A. The chart isomorphism identifies the localization of the original frame ring at $\det L$ with R_{chart}.

There is a natural homomorphism $\iota : R_0 \to R_{\mathrm{chart}}$. Define $\rho : R_{\mathrm{chart}} \to R_0$ by

$$\rho(\ell_{ij}) = \delta_{ij}, \qquad \rho(a_{ij}) = 0, \qquad \rho(u) = 1, \qquad \rho(q_a) = q_a, \quad \rho(c_{ai}) = c_{ai}.$$

This is well-defined because $u\det(\ell) - 1$ maps to $1 \cdot 1 - 1 = 0$. Moreover $\rho \circ \iota = \mathrm{id}_{R_0}$, so ι is injective. For the certified class $\bar{g} \neq 0$ of Lemma 2, we have

$$\iota(\bar{g})^2 = \iota(\bar{g}^2) = 0, \qquad \rho(\iota(\bar{g})) = \bar{g} \neq 0.$$

Thus $\iota(\bar{g})$ is a nonzero square-zero element of the chart ring. The crucial assertion is the split injection, not merely the existence of a surjective evaluation homomorphism.

In original frame coordinates the same class is

$$\widetilde{g} = g\big((2(L^{\mathsf{T}}B)_{aa})_{a=1}^{4}, L^{-1}P\big).$$

The evaluation above is induced by the explicit section

$$(q, C) \longmapsto \begin{bmatrix} I & C \\ Q/2 & -QC/2 \end{bmatrix}.$$

Its pairings satisfy the original equations because the only residual off-diagonal entries are those of $-C^{\mathsf{T}}QC$. Substituting this section into $\widetilde{g}$ recovers $g(q, C)$, so it cannot annihilate the certified class. Hence the localized frame ring is nonreduced. A localization of a reduced ring is reduced, proving nonreducedness of the full split frame ring.

For a general b, diagonalization and adjoining finitely many square roots give a finite separable extension L/K over which b becomes split. If the frame ring over K were reduced, its finite étale scalar extension to L would remain reduced, contradicting the split case. For an elementary justification of this preservation fact, embed a reduced finitely generated algebra in the finite product of the fraction fields of its minimal-prime quotients. Tensoring with L preserves this injection, and tensoring each of those fields with a finite separable extension gives a product of fields. The resulting algebra is reduced. □

The distinction between an open chart and a quotient is essential here: nonreducedness of a special fiber alone would not prove the theorem. Equation (3) supplies an open localization of the original frame scheme, with a regular inverse.

4 Stabilization and embedded components

Corollary 4. *For every $d \geq 8$ and $n \geq d + 1$, over every field of characteristic different from two and for every nondegenerate symmetric form b, the ideal $I_b(d, n)$ is not radical.*

Proof. For fixed dimension eight, adjoining further columns gives an injection of the nine-frame coordinate ring into the n-frame coordinate ring: setting the added columns equal to zero is a retraction. Thus Theorem 1 supplies the base case for every $n \geq 9$ and every nondegenerate form over K.

We now give the dimension step for a fixed quadratic space (E, b). Choose $e \in E$ with $a = b(e, e) \neq 0$ and put $W = e^{\perp}$. Such an e exists by polarization, since 2 is invertible and b is nondegenerate. Then $E = Ke \perp W$ and $b_W = b|_W$ is nondegenerate. We will prove

$$X_{b_W}(d-1, n-1) \text{ nonreduced} \quad \Longrightarrow \quad X_b(d, n) \text{ nonreduced.} \tag{6}$$

The fixed form b_W need not be split.

An explicit étale cover. Write v for the universal vector of E, $q = b(v, v)$, and $s = b(v, e)$. Let

$$A = K[E][q^{-1}], \qquad B = \operatorname{Spec} A, \qquad U = X_b(d, n)\big|_{b(\mathbf{v}_1, \mathbf{v}_1) \neq 0}.$$

Projection to the first vector defines $U \to B$. Define

$$S = \big(A[t]/(t^2 - q/a)\big)[(at - s)^{-1}], \qquad T = \operatorname{Spec} S. \tag{7}$$

The element t is a unit because $t^2 = q/a$ is a unit. The derivative $2t$ is therefore a unit. Consequently $T \to B$ is étale: it is an open subscheme of the finite étale double cover defined by $t^2 = q/a$. It is also surjective. Over a geometric point of B the two roots are distinct and nonzero, and $at - s$ cannot vanish at both roots. In particular, T is nonempty; more explicitly, $(v, t) = (e, -1)$ is a K-point, with $at - s = -2a \neq 0$. The standard étale criterion and its stability under base change are recalled in [4].

A regular isometry trivializing the complement. Over S set

$$w = v - te, \qquad \gamma = b(w, w) = 2t(at - s) \in S^\times, \qquad r_w(z) = z - \frac{2b(z, w)}{\gamma} w.$$

Direct expansion gives

$$r_w^2 = 1, \qquad b(r_w z, r_w z') = b(z, z'), \qquad r_w(v) = te.$$

For the last identity, $b(v, w) = q - ts = t(at - s)$, so the coefficient subtracted from v is exactly one. Thus r_w identifies the universal orthogonal complement $v^\perp$ with the constant quadratic module $W \otimes_K S$. All its coefficients are regular on T, since γ is a unit.

It follows that there is an isomorphism of T-schemes

$$U \times_B T \;\simeq\; T \times_K X_{b_W}(d - 1, n - 1). \tag{8}$$

Explicitly, for any S-algebra C, it sends a frame $(v, z_2, \ldots, z_n)$ to $(r_w z_2, \ldots, r_w z_n)$, retaining the base point (v, t). Indeed, $b(v, z_i) = 0$ gives $t\, b(e, r_w z_i) = 0$, and t is a unit; the remaining pairings are preserved by the isometry. Conversely, a frame $(y_2, \ldots, y_n)$ in $W \otimes_K C$ is sent to $(v, r_w y_2, \ldots, r_w y_n)$. This inverse lands in the original frame scheme because

$$b(v, r_w y_i) = b(r_w v, y_i) = t\, b(e, y_i) = 0, \qquad b(r_w y_i, r_w y_j) = b(y_i, y_j) = 0.$$

Both compositions fix (v, t) and every remaining vector, since $r_w(r_w z) = z$. These formulas are natural for arbitrary C, including nonreduced algebras, and prove (8) scheme-theoretically.

Transport of nonreducedness. Let D be the coordinate ring of $X_{b_W}(d - 1, n - 1)$ and suppose it contains a nonzero nilpotent η. Evaluation at $(e, -1)$ defines $\epsilon : S \to K$: it sends q to a, s to a, and t to -1. The defining quadratic maps to zero; the inverted elements q and $at - s$ map to a and $-2a$, both nonzero. Thus ϵ is well-defined. The maps

$$\jmath : D \to S \otimes_K D, \quad d \mapsto 1 \otimes d, \qquad \rho_S(s \otimes d) = \epsilon(s) d$$

satisfy $\rho_S \circ \jmath = \mathrm{id}_D$. If $\jmath(\eta) = 0$, applying ρ_S would give $\eta = 0$, a contradiction. Also $\jmath(\eta)^N = \jmath(\eta^N) = 0$ whenever $\eta^N = 0$. Hence $1 \otimes \eta$ is a nonzero nilpotent. By (8), $U \times_B T$ is nonreduced. If $X_b(d, n)$ were reduced, its open subscheme U would be reduced, and the étale base change $U \times_B T \to U$ would be reduced as well, a contradiction [4].

In this specific construction the last preservation fact also has an elementary algebraic proof. If $R = \Gamma(U, \mathcal{O}_U)$ is reduced, embed it in the finite product of the fraction fields of its minimal-prime quotients. Since $t^2 - q/a$ is monic, its quotient is free of rank two over R and embeds in the corresponding product of quadratic algebras over those fields. Each quadratic is separable because q/a is a unit and 2 is invertible; its algebra is a product of fields. Thus $R[t]/(t^2 - q/a)$ is reduced, and localization at $at - s$ remains reduced. This proves (6).

Finally, induct on d, with the assertion made for every nondegenerate form over the given field. For $d > 8$ and $n \geq d + 1$, the complement W has dimension $d - 1$ and $n - 1 \geq (d - 1) + 1$, so the induction hypothesis applies to b_W. Equation (6) completes the proof directly over K. □

Casabella–Sammartano prove that the frame schemes are generically reduced [2, Theorem 7]. Combining that imported theorem with Corollary 4 shows that these frame rings have embedded associated primes. Indeed, a nonzero nilradical has an associated prime of the ambient ring, while generic reducedness excludes minimal primes from its support. The imported generic-reducedness theorem is not needed for our nonradicality proof. We do not identify all embedded primes.

4.1 Minimality and remaining parameter ranges

Throughout this discussion $\operatorname{char} K \neq 2$ and b is nondegenerate. The pair $(8, 9)$ is a certified counterexample; we do not prove it is minimal, either in ambient dimension or in number of vectors.

There are substantial known radical ranges. For $d = 1$, the ideal is generated by squarefree monomials. For $d = 2$, radicality for every n follows from [3, Theorem 1.1]; for arbitrary b, first pass to a finite separable extension making the form standard, and then descend reducedness. For $d \geq 2$, the complete-intersection theorem of Casabella–Sammartano [2, Theorem 7.5] gives radicality for $n \leq N_{\mathrm{CI}}(d)$, where its bound, solved for n, is

$$N_{\mathrm{CI}}(d) = \begin{cases} \left\lfloor (d+1+\sqrt{4d+1})/2 \right\rfloor, & d \text{ even}, \\ \left\lfloor (d+2+\sqrt{4d-3})/2 \right\rfloor, & d \text{ odd}. \end{cases}$$

Combining these results with Corollary 4 gives the following table. "Unresolved" means not decided by these cited theorems and the present paper; failure of the complete-intersection criterion alone does not imply nonradicality.

d	Known radical	Nonradical here	Unresolved
$1, 2$	all n	—	none
$3, 4$	$n \leq 4$	—	$n \geq 5$
5	$n \leq 5$	—	$n \geq 6$
6	$n \leq 6$	—	$n \geq 7$
7	$n \leq 7$	—	$n \geq 8$
8	$n \leq 7$	$n \geq 9$	$n = 8$
9	$n \leq 8$	$n \geq 10$	$n = 9$
10	$n \leq 8$	$n \geq 11$	$n = 9, 10$
$d \geq 11$	$n \leq N_{\mathrm{CI}}(d)$	$n \geq d+1$	$N_{\mathrm{CI}}(d) < n \leq d$

In particular, for fixed dimension eight the least number of vectors giving a nonradical ideal is either eight or nine: $(8, 8)$ is the only remaining case in that dimension. This does not settle minimality across dimensions. Even over $\mathbb{C}$, the results above only place the least ambient dimension between three and eight, and the least number of vectors between five and nine, with these two minima taken separately. The smaller pairs $(3, 5)$ and $(4, 5)$ are already outside the ranges settled here. Finally, the propagation used in Corollary 4 adds columns or increases both d and n by one; it supplies no converse that would settle these remaining cases.

5 Certificate format and reproducibility

The accompanying package contains the exact proof data and a read-only checker. The principal files are:

File	Purpose
`certificate.json`	Integer witness, membership identity and dual functional.
`check_certificate.py`	Exact expansion and exhaustive dual verification.
`THEOREM.txt`	Complete written chart, descent and stabilization argument.
`replay/viewer.html`	Public research events, graph revisions and pinned evidence.

The short name `certificate.json` is a byte-for-byte copy of `diagonal_integral_certificate.json`. Its SHA-256 digest, split across two lines for readability, is

```
d0e1079fee0df2878e011628c1bb64330
4a3cacb0b097c8393b5b347da8c5c69
```

Each monomial is encoded by a list of 24 nonnegative exponents in the order specified above. The `g` and `dual` lists contain pairs `[exponents,coefficient]`. A membership record has an `edge` (zero-based column indices), a monomial `multiplier`, and an integer `coefficient`. It represents that coefficient times the multiplier times the corresponding original F_{ij}.

From the package directory, run

```
python -X utf8 check_certificate.py certificate.json
python -X utf8 replay/verify.py --compute
```

Both commands use only the Python standard library. The first verifies the algebraic certificate directly. The second additionally checks the packet's hashes, event chain, graph deltas and source mapping, and runs a fresh certificate verification in a temporary directory. The arithmetic check is independent of the exploration software.

The core of the dual check can be expressed without any elimination:

```
for M, value in dual_support:
    for edge, generator in original_generators:
        for T in monomials(generator):
            if T divides M:
                sums[edge, M/T] += value
assert every entry of sums is zero
assert sum(g[M] * dual[M] for M in g) == 1
```

All generator coefficients are one. This checks every generator multiple capable of pairing nontrivially with the functional; it is not a random test. The full checker also expands the displayed witness and the membership identity, and verifies the square-zero identity.

6 Discovery and status

The counterexample was found during an atlas-guided exploration of parameter torsion in the diagonal chart. A Gröbner calculation over $\mathbb{F}_3$ produced a polynomial divisible by q_4^2, whose quotient by one factor of q_4 was a nonmembership candidate. An initial run exhausted its memory allowance while serializing the basis; streaming the completed basis recovered the candidate. The retained discovery output has 8,525 basis elements. None of that basis is trusted by the final certificate checker.

Independent fine-graded linear algebra over $\mathbb{F}_3$ and $\mathbb{F}_{32003}$ recovered a membership identity and a separating functional. Lifting the latter coefficients to small signed integers suggested the integral certificate. Direct integer expansion, including the exhaustive dual test, then established Lemma 2. This last step is the proof-producing step; neither modular agreement nor floating-point evidence is used as a substitute.

This draft supplies a computer-assisted proof and replayable exact data. It is not a formal proof-assistant development. A bounded primary-source comparison found the conjecture still stated in the October 30, 2026 version of [2]; historical priority remains provisional, and independent review beyond the author's initial review is not recorded. Earlier general-graph examples do not answer the complete-graph conjecture addressed here.

7 A visual guide to the reasoning route

Figures 1–3 show the three tabs of the accompanying mathematical route map. They give complementary views of the RRE exploration [5] that led to the certificate and its geometric interpretation. The stations are arranged in logical order; this is a retrospective guide to the argument, rather than a chronological transcript of the computation. The event replay described in Section 5 retains the computational history and pinned evidence.

The *All math routes* view places the successful route beside the earlier attempts. The two inherited obligations, origin control and pencil torsion, belonged to a joint induction plan and did not close. The smaller-chart computations supply evidence only over the tested finite field; their passing results do not decide whether the characteristic-zero witness survives that localization. The *Proof route* view emphasizes the premises of the finished argument: the two integral certificates, the chart isomorphism with its split injection, separable descent, and stabilization. In particular, the finite-field candidate is not a premise of Lemma 2. The certificate applies in every characteristic; the chart transport and frame statements require characteristic different from two.

The *Provenance* view distinguishes published inputs from context recorded in the earlier RRE run. Its inherited labels report that earlier record; they are not additional radicality results established or used in this paper, and do not enlarge the ranges in Section 4.1. Generic reducedness is the separate literature input used for the embedded-prime consequence, not for Theorem 1. The mathematical claims rest on the proofs and certificate checks above.

The screenshots retain each active tab, the legend, and the full map; the surrounding introduction and station-detail panel are omitted. The interactive version is supplied as `replay/RRE012_math_route_map.html` in the accompanying package.

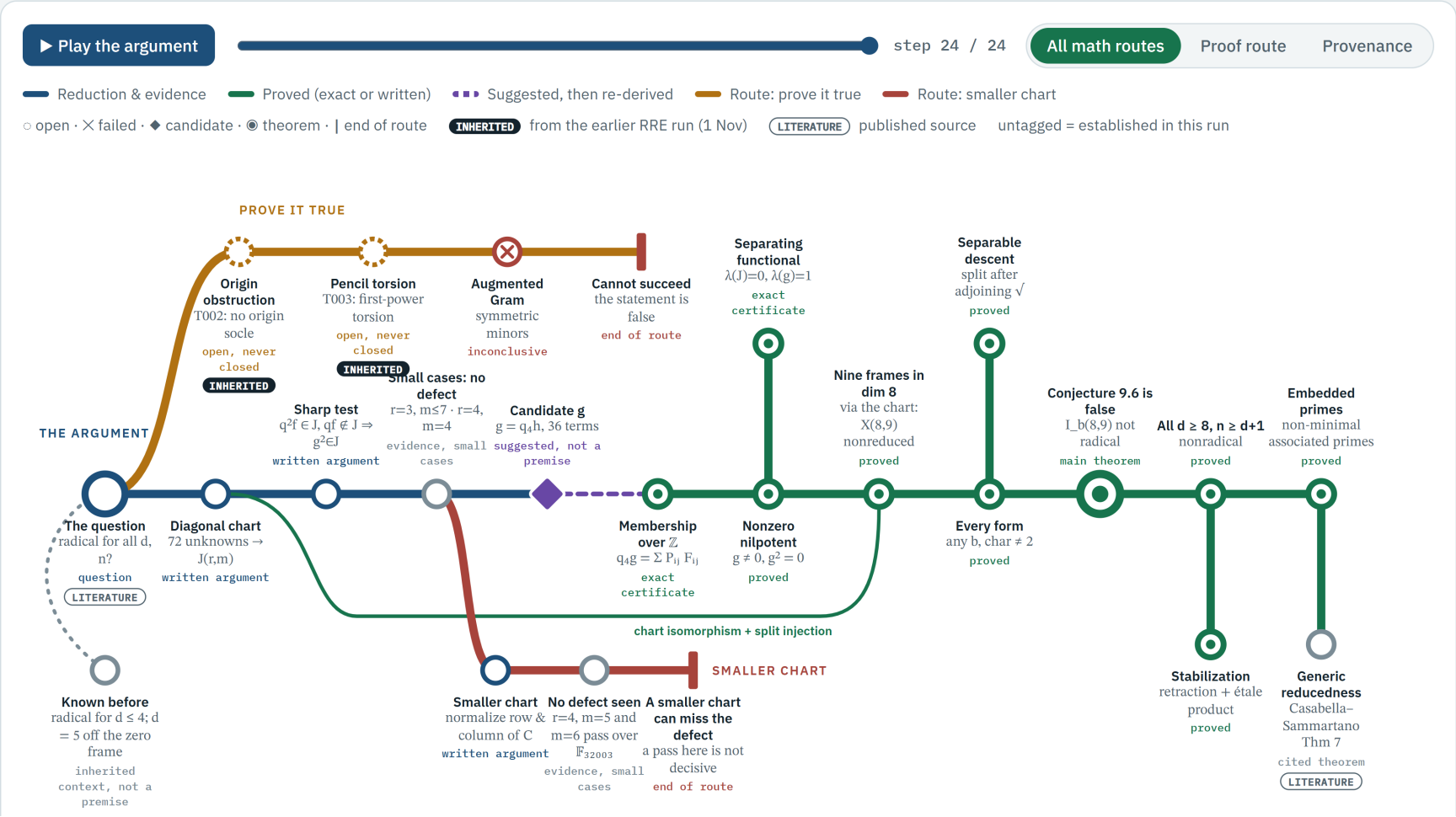


Figure 1: **All math routes.** The complete logical route from the radicality question to the counterexample and its consequences. Side branches record the unsuccessful induction plan and the inconclusive smaller-chart tests. The violet candidate is a discovery aid; the green route is supported by exact certificates and written proofs.

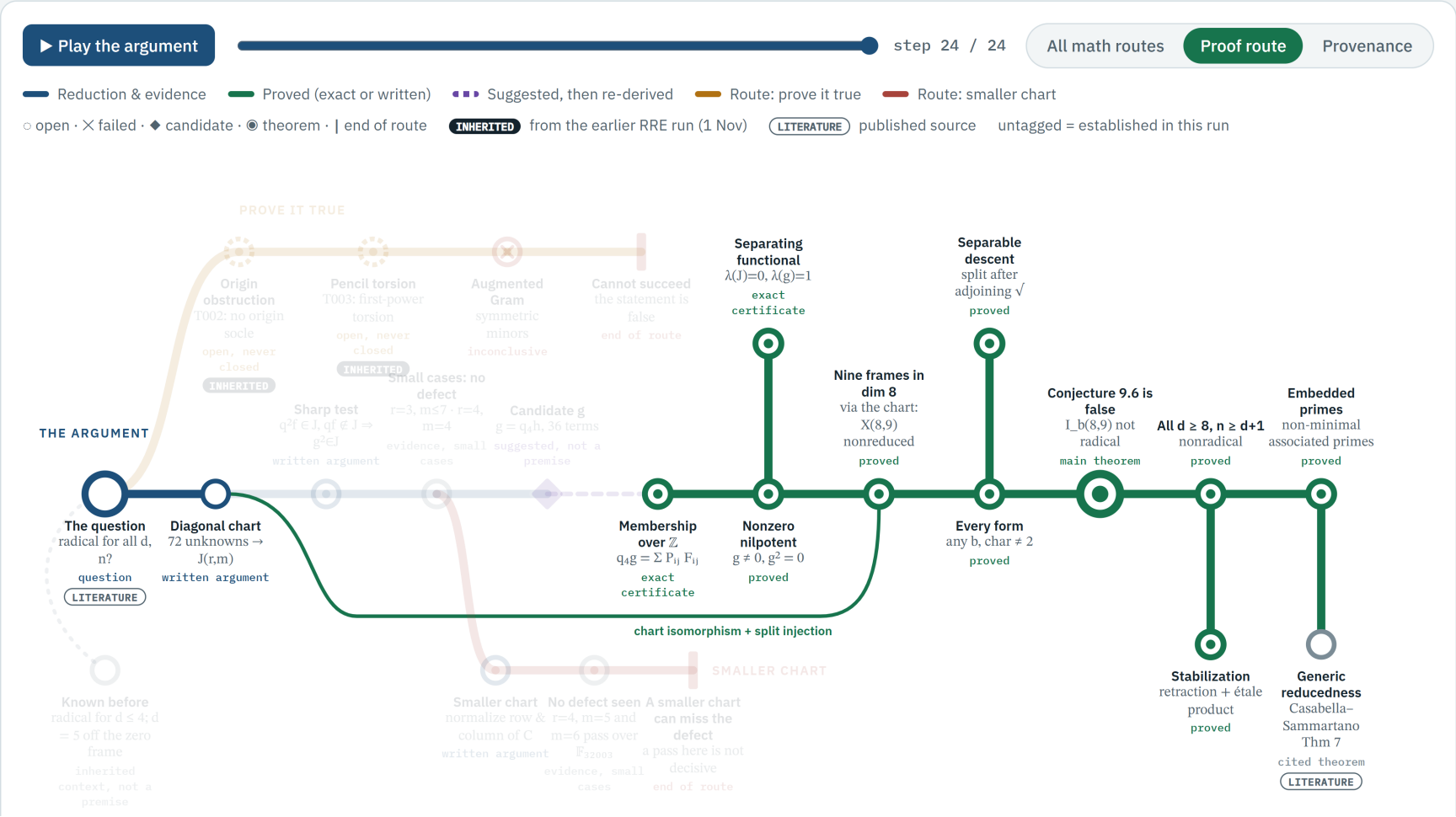


Figure 2: **Proof route.** Exploratory stations are dimmed. The membership identity and separating functional jointly certify the nonzero nilpotent. The lower arc makes explicit the additional chart isomorphism and split injection needed to transport it to the frame scheme. Descent, stabilization, and the cited generic-reducedness theorem enter at their respective steps.

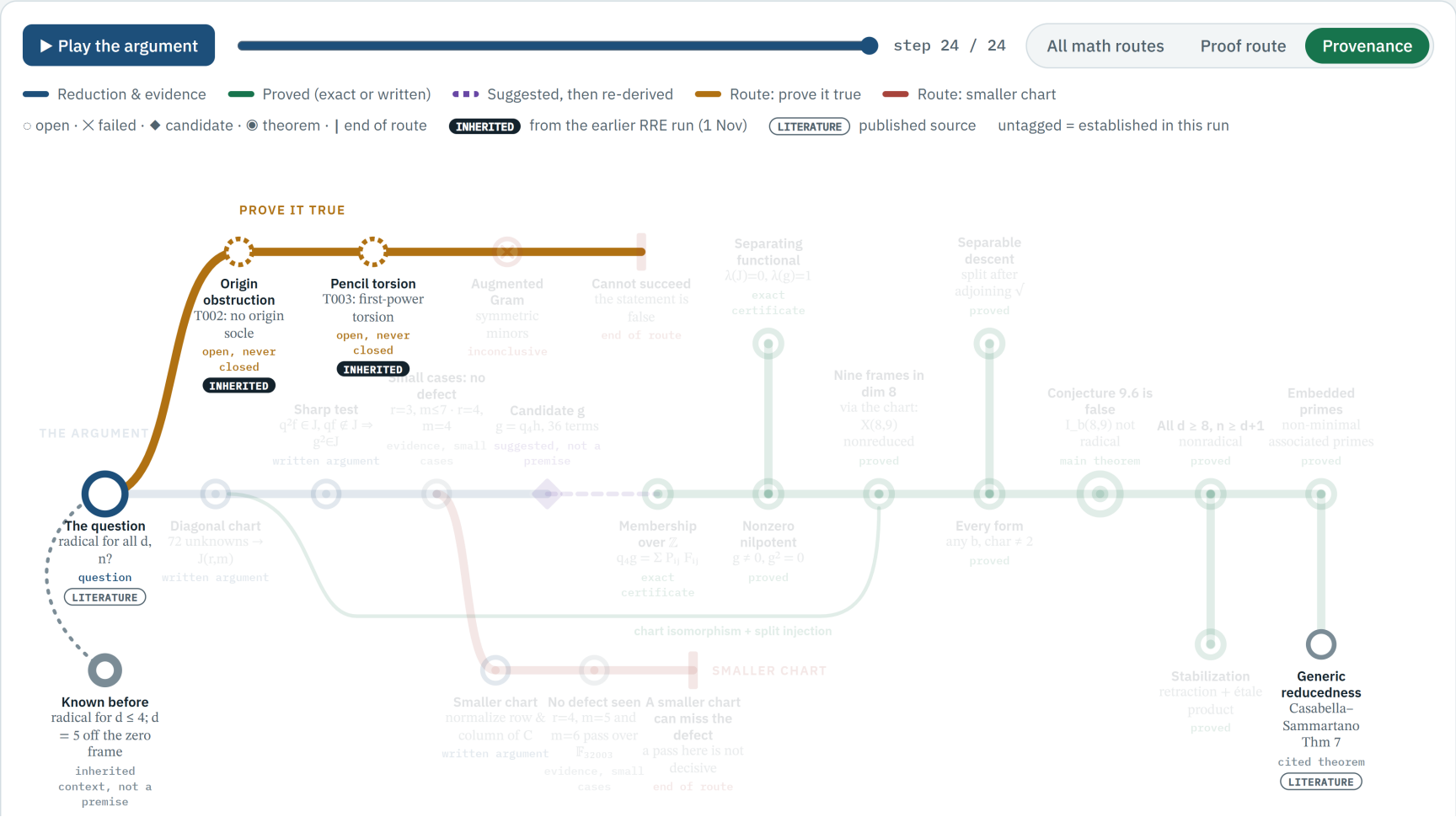


Figure 3: **Provenance.** Tags distinguish literature inputs from inherited research context; the other stations are dimmed. The characteristic-zero small-dimensional claims in the inherited context are historical labels, not premises of this paper. The conjecture supplies the question, while generic reducedness is used only for the embedded-prime consequence.

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